Electrochemistry: Master Comprehensive Tutorial

1. Learning Objectives & Exam Relevance

In the NEET UG examination, **Electrochemistry** is one of the highest-weightage topics in Physical Chemistry, typically contributing 2 to 3 direct questions every year. Problems primarily test conceptual clarity, standard thermodynamic conversions, Nernst equation calculations, conductance interpretations, Kohlrausch's law applications, and the chemical reactions underlying commercial cells and corrosion.

Key NEET UG Focus Areas

- **Galvanic vs. Electrolytic Cells:** Daniell cell operation, the effect of external opposing voltage ($E_{\text{ext}} < 1.1\ \text{V}$, $E_{\text{ext}} = 1.1\ \text{V}$, and $E_{\text{ext}} > 1.1\ \text{V}$).
- **Standard Electrode Potential:** IUPAC reduction convention, Standard Hydrogen Electrode (SHE), electrochemical series predictions.
- **Nernst Equation:** Cell potential dependence on concentration and temperature, calculation of equilibrium constant (K_c) and standard Gibbs free energy change ($\Delta_r G^\circ$).
- **Electrolytic Conductance:** Specific conductivity (κ), molar conductivity (Λ_m), cell constant (G^*), and dilution effects.
- **Kohlrausch's Law:** Independent migration of ions, calculation of limiting molar conductivity (Λ_m°) for weak electrolytes, determination of degree of dissociation (α) and dissociation constant (K_a).
- **Electrolysis:** Faraday's 1st and 2nd laws of electrolysis, preferential discharge theory, and overpotential.
- **Batteries & Corrosion:** Anode and cathode reactions in Leclanché cells, Mercury cells, Lead storage batteries, Fuel cells (H_2–O_2), and the electrochemical mechanism of rust formation.

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2. Electrochemical Cells & Daniell Cell

An **electrochemical cell** consists of two metallic electrodes dipping into electrolytic solutions. These cells fall into two distinct thermodynamic regimes:

1. **Galvanic (Voltaic) Cell:** A device that converts the chemical Gibbs free energy of a spontaneous redox reaction ($\Delta G < 0$) into usable electrical work (e.g., Daniell cell).
2. **Electrolytic Cell:** A device that utilizes electrical energy from an external power source to drive a non-spontaneous chemical transformation ($\Delta G > 0$).

2.1 The Daniell Cell Construction and Principle

The Daniell cell utilizes the spontaneous redox reaction between metallic zinc and aqueous copper(II) ions:

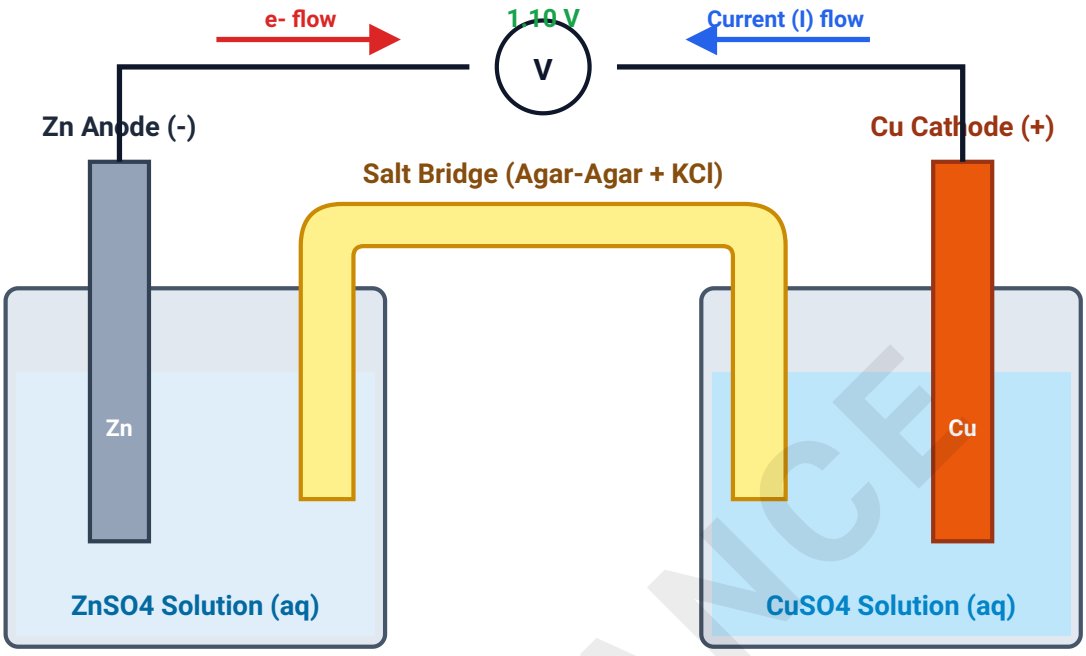
$$\text{Zn(s)} + \text{Cu}^{2+}\text{(aq)} \longrightarrow \text{Zn}^{2+}\text{(aq)} + \text{Cu(s)}$$

This overall reaction is partitioned into two spatially segregated half-cells:

- **Anode (Oxidation Half-Cell):** Zinc rod immersed in aqueous ZnSO_4. Oxidation occurs here:
  $$\text{Zn(s)} \longrightarrow \text{Zn}^{2+}\text{(aq)} + 2\text{e}^-$$
  Because electrons accumulate at this electrode before entering the external circuit, the anode in a galvanic cell carries a **negative (-) polarity**.
- **Cathode (Reduction Half-Cell):** Copper rod immersed in aqueous CuSO_4. Reduction occurs here:
  $$\text{Cu}^{2+}\text{(aq)} + 2\text{e}^- \longrightarrow \text{Cu(s)}$$

Cations from the solution consume electrons at this surface, giving the cathode a **positive (+) polarity**.

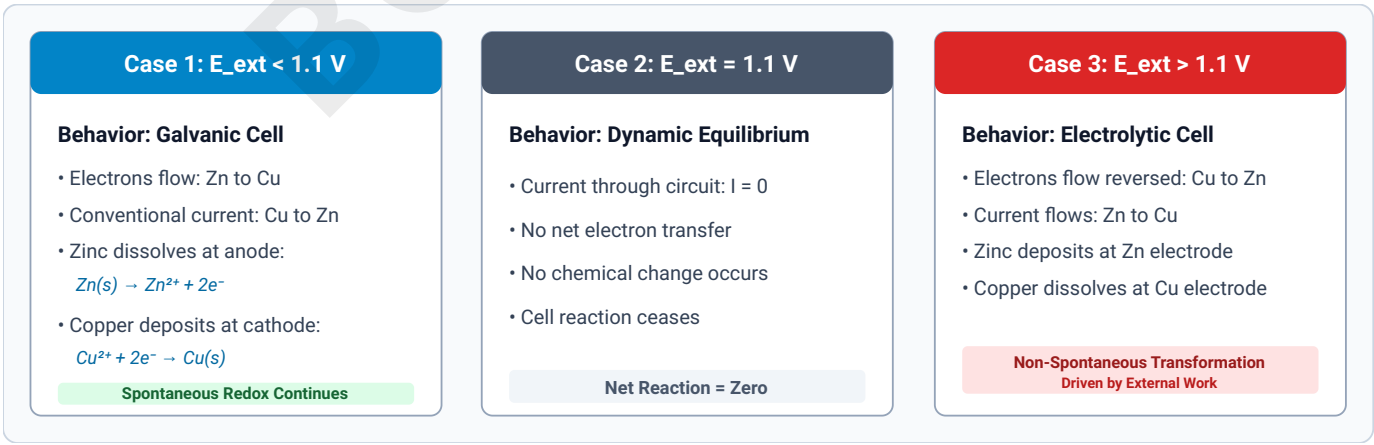
- **Salt Bridge:** An inverted U-tube filled with an inert electrolyte agar-agar gel (e.g., KCl, KNO_3, or NH_4NO_3) whose ionic mobilities are nearly identical. It completes the internal electrical circuit and maintains electrical neutrality in both half-cell solutions without allowing physical mixing of the bulk liquids.



Tutorial Diagram 1: Daniell cell showing external electron flow (Zn to Cu), conventional current (Cu to Zn), and internal ionic migration via the salt bridge.

2.2 Influence of External Opposing Potential (E_{ext})

When an opposing external potential (E_{ext}) is connected across the Daniell cell and gradually raised, three distinct regimes are observed:



Tutorial Diagram 2: Operational modes of Daniell cell as external opposing potential increases across the threshold value of 1.10 V.

2.3 Cell Notation & Electromotive Force (EMF)

According to IUPAC conventions:

- The **anode** (oxidation half-cell) is written on the **left-hand side**.

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- The **cathode** (reduction half-cell) is written on the **right-hand side**.
- A single vertical line ($|$) represents a phase boundary (e.g., between solid metal and aqueous solution).
- A double vertical line ($||$) indicates a salt bridge connecting the two electrolyte compartments.

For the Daniell cell:

$$\mathrm{Zn(s)} \mid \mathrm{Zn^{2+}(aq)} \parallel \mathrm{Cu^{2+}(aq)} \mid \mathrm{Cu(s)}$$

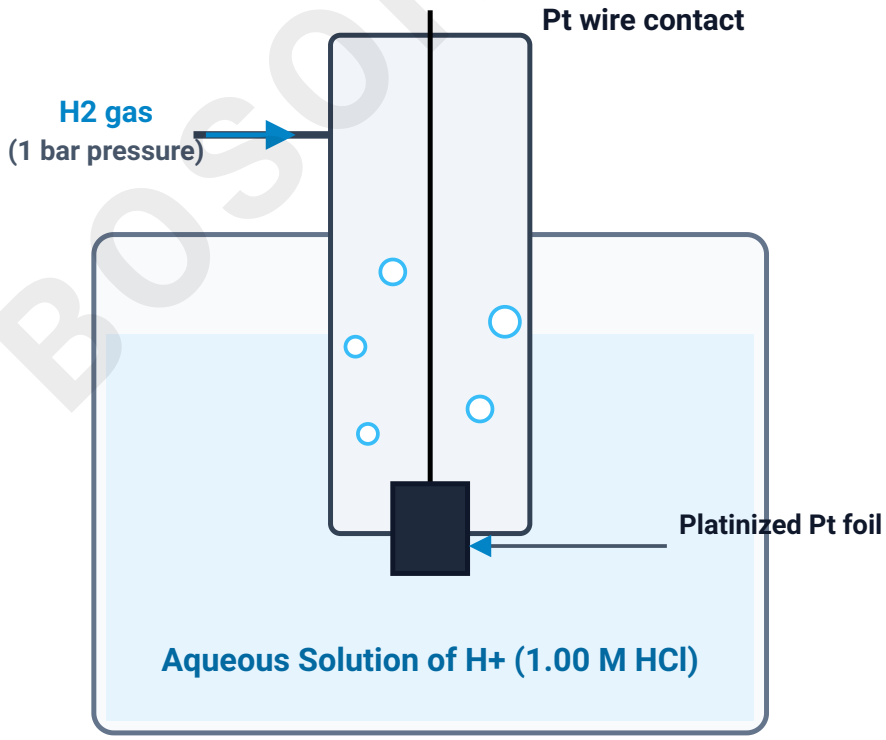
The cell potential when zero net current is drawn is the **electromotive force (EMF)**:

$$E_{\text{cell}} = E_{\text{cathode}} - E_{\text{anode}} = E_{\text{right}} - E_{\text{left}}$$

CRITICAL IUPAC RULE FOR NEET UG: Always use **Standard Reduction Potentials** for both E_{right} and E_{left}. Never add oxidation potential and reduction potential when using the formula $E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}}$.

3. Standard Hydrogen Electrode (SHE) & Electrochemical Series

The absolute potential of an isolated half-cell cannot be measured experimentally because oxidation and reduction cannot proceed in isolation. Only potential differences between two electrodes can be measured. Therefore, a reference electrode is chosen by convention: the **Standard Hydrogen Electrode (SHE)**, assigned an arbitrary potential of exactly $0.00\ \mathrm{V}$ at all temperatures.



Tutorial Diagram 3: Standard Hydrogen Electrode (Pt(s) | H_2(g, 1 bar) | H^+(aq, 1 M)), assigned an arbitrary reference potential of 0.00 V.

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3.1 Measurement of Standard Electrode Potentials

To measure the standard electrode potential E° of any couple $\mathrm{M}^{n+}/\mathrm{M}$, a cell is constructed with the SHE as the left electrode (anode) and the unknown half-cell as the right electrode (cathode):

$$\mathrm{Pt(s)} \mid \mathrm{H_2(g, 1\,bar)} \mid \mathrm{H^+(aq, 1\,M)} \parallel \mathrm{M^{n+}(aq, 1\,M)} \mid \mathrm{M(s)}$$

The measured cell EMF yields:

$$E^\circ_{\text{cell}} = E^\circ_{\text{R}} - E^\circ_{\text{L}} = E^\circ_{\mathrm{M}^{n+}/\mathrm{M}} - 0 = E^\circ_{\mathrm{M}^{n+}/\mathrm{M}}$$

- For $\mathrm{Cu}^{2+}/\mathrm{Cu}$: $E^\circ_{\text{cell}} = +0.34\,\mathrm{V}$. The positive sign indicates that Cu^{2+} ions are reduced more readily than H^+ ions. Thus, copper does not displace hydrogen gas from non-oxidizing dilute acids (e.g., HCl).
- For $\mathrm{Zn}^{2+}/\mathrm{Zn}$: $E^\circ_{\text{cell}} = -0.76\,\mathrm{V}$. The negative sign indicates that zinc can reduce H^+ ions to $\mathrm{H_2}$ gas, dissolving readily in dilute mineral acids.

3.2 Standard Reduction Potentials at 298 K

Half-Reaction (Reduction)	E° (Volts)	Oxidizing / Reducing Power Interpretation
$\mathrm{F_2(g) + 2e^- \longrightarrow 2F^-}$	**+2.87**	$\mathrm{F_2}$ is the *strongest oxidizing agent*; $\mathrm{F^-}$ is the weakest reducing agent.
$\mathrm{Co^{3+} + e^- \longrightarrow Co^{2+}}$	+1.81	Strong oxidizing agent.
$\mathrm{MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O}$	+1.51	Potent oxidizing agent in acidic medium.
$\mathrm{Cl_2(g) + 2e^- \longrightarrow 2Cl^-}$	+1.36	Oxidizes $\mathrm{Br^-}$ and $\mathrm{I^-}$.
$\mathrm{Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O}$	+1.33	Standard analytical oxidizing agent.
$\mathrm{O_2(g) + 4H^+ + 4e^- \longrightarrow 2H_2O}$	+1.23	Thermodynamic water oxidation potential.
$\mathrm{Br_2 + 2e^- \longrightarrow 2Br^-}$	+1.09	Oxidizes $\mathrm{I^-}$ to $\mathrm{I_2}$.
$\mathrm{Ag^+ + e^- \longrightarrow Ag(s)}$	+0.80	Easily reduced by Cu.
$\mathrm{Fe^{3+} + e^- \longrightarrow Fe^{2+}}$	+0.77	Can oxidize $\mathrm{I^-}$, but cannot oxidize $\mathrm{Br^-}$.
$\mathrm{I_2 + 2e^- \longrightarrow 2I^-}$	+0.54	Mild oxidizing agent.
$\mathrm{Cu^{2+} + 2e^- \longrightarrow Cu(s)}$	+0.34	Positive E°; does not release $\mathrm{H_2}$ with dilute HCl.
$\mathrm{2H^+(aq) + 2e^- \longrightarrow H_2(g)}$	**0.00**	**Arbitrary standard reference point (SHE).**
$\mathrm{Pb^{2+} + 2e^- \longrightarrow Pb(s)}$	-0.13	Releases $\mathrm{H_2}$ from acids slowly.
$\mathrm{Sn^{2+} + 2e^- \longrightarrow Sn(s)}$	-0.14	Negative reduction potential.
$\mathrm{Ni^{2+} + 2e^- \longrightarrow Ni(s)}$	-0.25	Displaces hydrogen from acids.
$\mathrm{Fe^{2+} + 2e^- \longrightarrow Fe(s)}$	-0.44	Susceptible to corrosion.
$\mathrm{Zn^{2+} + 2e^- \longrightarrow Zn(s)}$	-0.76	Standard sacrificial coating metal (galvanization).
$\mathrm{Al^{3+} + 3e^- \longrightarrow Al(s)}$	-1.66	Extracted by fused-salt electrolysis.
$\mathrm{Mg^{2+} + 2e^- \longrightarrow Mg(s)}$	-2.36	Strong reducing agent.
$\mathrm{Na^+ + e^- \longrightarrow Na(s)}$	-2.71	Powerful reducing metal.
$\mathrm{Li^+ + e^- \longrightarrow Li(s)}$	**-3.05**	$\mathrm{Li(s)}$ is the *strongest reducing agent* in aqueous solution due to highest hydration energy of $\mathrm{Li^+}$.

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4. The Nernst Equation & Electrochemical Thermodynamics

4.1 Mathematical Formulation

Electrode potentials depend on concentration and temperature. For a general reduction half-cell:

$$\mathrm{M}^{n+}(\mathrm{aq}) + n\mathrm{e}^- \longrightarrow \mathrm{M}(\mathrm{s})$$

The Nernst equation states:

$$E_{(\mathrm{M}^{n+}/\mathrm{M})} = E^\circ_{(\mathrm{M}^{n+}/\mathrm{M})} - \frac{RT}{nF}\ln\frac{[\mathrm{M(s)}]}{[\mathrm{M}^{n+}(\mathrm{aq})]}$$

By thermodynamic convention, the active mass (activity) of a pure solid or liquid is taken as unity ($[\mathrm{M(s)}] = 1$). Substituting $R = 8.314\,\mathrm{J\cdot K^{-1}\cdot mol^{-1}}$, $F = 96487\,\mathrm{C\cdot mol^{-1}}$, and $T = 298.15\,\mathrm{K}$, with $\ln x = 2.303\log_{10} x$:

$$\frac{2.303 \times 8.314 \times 298.15}{96487} \approx 0.05916\,\mathrm{V} \approx \mathbf{0.059\,V}$$

$$E_{(\mathrm{M}^{n+}/\mathrm{M})} = E^\circ_{(\mathrm{M}^{n+}/\mathrm{M})} - \frac{0.059}{n}\log\frac{1}{[\mathrm{M}^{n+}]}$$

4.2 Complete Cell Reaction

For any generalized redox reaction involving n moles of electrons:

$$a\mathrm{A} + b\mathrm{B} \xrightarrow{n\mathrm{e}^-} c\mathrm{C} + d\mathrm{D}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.059}{n}\log Q$$

$$\text{where } Q = \frac{[\mathrm{C}]^c[\mathrm{D}]^d}{[\mathrm{A}]^a[\mathrm{B}]^b}$$

4.3 Equilibrium Constant (K_c) from the Nernst Equation

As a galvanic cell operates, reactants are consumed and products accumulate. The cell potential decreases steadily until dynamic chemical equilibrium is established. At equilibrium:

$$E_{\text{cell}} = 0 \quad \text{and} \quad Q = K_c$$

$$0 = E^\circ_{\text{cell}} - \frac{2.303RT}{nF}\log K_c$$

$$\mathbf{E^\circ_{cell}} = \frac{\mathbf{0.059}}{\mathbf{n}}\log \mathbf{K_c} \quad (\text{at } 298\,\mathrm{K})$$

4.4 Cell Potential and Gibbs Free Energy ($\Delta_r G$)

The maximum reversible electrical work done by a galvanic cell equals the decrease in the system's Gibbs free energy:

$$W_{\text{elec, max}} = -\Delta_r G = nFE_{\text{cell}}$$

$$\mathbf{\Delta_r G = -nFE_{cell}}$$

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$$\boldsymbol{\Delta_r G^\circ = -nFE^\circ_{cell} = -RT\ln K_c}$$

Thermodynamic Properties in NEET:

- E_{cell} is an **intensive property**; its numerical value does not change when the stoichiometric coefficients are multiplied by a factor.
- $\Delta_r G$ is an **extensive property**; multiplying a reaction by 2 doubles both n and the magnitude of $\Delta_r G$.
- Spontaneity criteria: For a spontaneous forward process, $\Delta_r G < 0$, which requires $E_{\text{cell}} > 0$.

5. Conductance of Electrolytic Solutions

5.1 Fundamental Parameters & SI Units

Parameter	Symbol	Governing Formula	SI Unit	Common Unit
Electrical Resistance	R	$R = \rho\frac{l}{A}$	Ω (ohm)	Ω
Resistivity	ρ	$\rho = R\frac{A}{l}$	$\Omega \cdot \text{m}$	$\Omega \cdot \text{cm}$ ($1\,\Omega \cdot \text{m} = 100\,\Omega \cdot \text{cm}$)
Conductance	G	$G = \frac{1}{R}$	S (siemens) $= \Omega^{-1}$	mho or Ω^{-1}
Conductivity (Specific Conductance)	κ	$\kappa = \frac{1}{\rho} = G\left(\frac{l}{A}\right)$	$\text{S} \cdot \text{m}^{-1}$	$\text{S} \cdot \text{cm}^{-1}$ ($1\,\text{S} \cdot \text{cm}^{-1} = 100\,\text{S} \cdot \text{m}^{-1}$)
Cell Constant	G^*	$G^* = \frac{l}{A} = R\kappa$	m^{-1}	cm^{-1} ($1\,\text{cm}^{-1} = 100\,\text{m}^{-1}$)
Molar Conductivity	Λ_m	$\Lambda_m = \frac{\kappa}{c}$	$\text{S} \cdot \text{m}^2 \cdot \text{mol}^{-1}$	$\text{S} \cdot \text{cm}^2 \cdot \text{mol}^{-1}$

5.2 Conversion Formulas for Molar Conductivity

$$\boldsymbol{\Lambda_m\,(\text{S}\cdot\text{cm}^2\cdot\text{mol}^{-1}) = \frac{\kappa\,(\text{S}\cdot\text{cm}^{-1})\times 1000}{\text{Molarity}\,(\text{mol}\cdot\text{L}^{-1})}}$$

$$\boldsymbol{\Lambda_m\,(\text{S}\cdot\text{m}^2\cdot\text{mol}^{-1}) = \frac{\kappa\,(\text{S}\cdot\text{m}^{-1})}{c\,(\text{mol}\cdot\text{m}^{-3})} = \frac{\kappa\,(\text{S}\cdot\text{m}^{-1})}{1000\times\text{Molarity}\,(\text{mol}\cdot\text{L}^{-1})}}$$

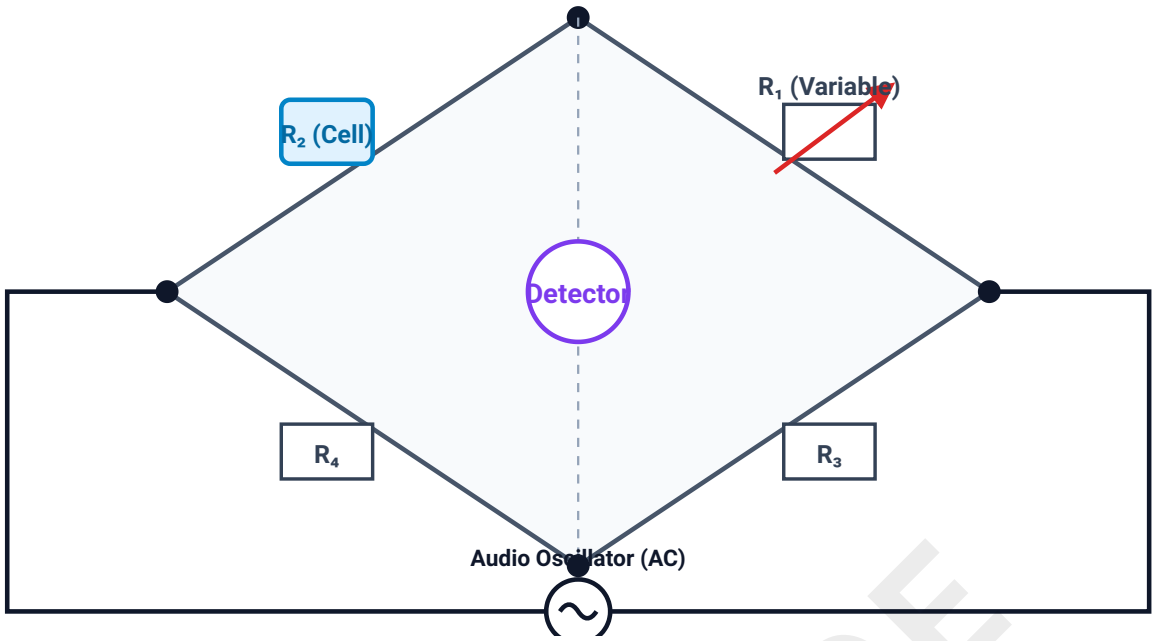
$$1\,\text{S}\cdot\text{m}^2\cdot\text{mol}^{-1} = 10^4\,\text{S}\cdot\text{cm}^2\cdot\text{mol}^{-1} \iff 1\,\text{S}\cdot\text{cm}^2\cdot\text{mol}^{-1} = 10^{-4}\,\text{S}\cdot\text{m}^2\cdot\text{mol}^{-1}$$

5.3 Experimental Measurement: AC Bridge & Conductivity Cell

Accurate measurement of the electrical resistance of an electrolytic solution requires addressing two key experimental hurdles:

1. **Polarization Error:** Direct current (DC) causes continuous electrolysis, which alters the ionic composition near the electrodes. This is eliminated by using an **alternating current (AC)** source operating in the audio-frequency range (550 to 5000 Hz).
2. **Connection Problem:** A liquid solution cannot be attached with simple alligator clips. A specially designed glass **conductivity cell** with platinized platinum electrodes of defined geometry is used.

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Tutorial Diagram 4: Wheatstone bridge configuration for measuring electrolytic solution resistance using an audio-frequency AC oscillator. At balance: $R_2 = \frac{R_1 R_4}{R_3}$.

6. Variation of κ and Λ_m with Concentration & Kohlrausch's Law

6.1 Why Conductivity (κ) Always Decreases on Dilution

> **Conductivity (κ)** is physically defined as the conductance of a unit volume ($1\ \text{cm}^3$ or $1\ \text{m}^3$) of solution enclosed between electrodes of unit cross-sectional area separated by unit length.

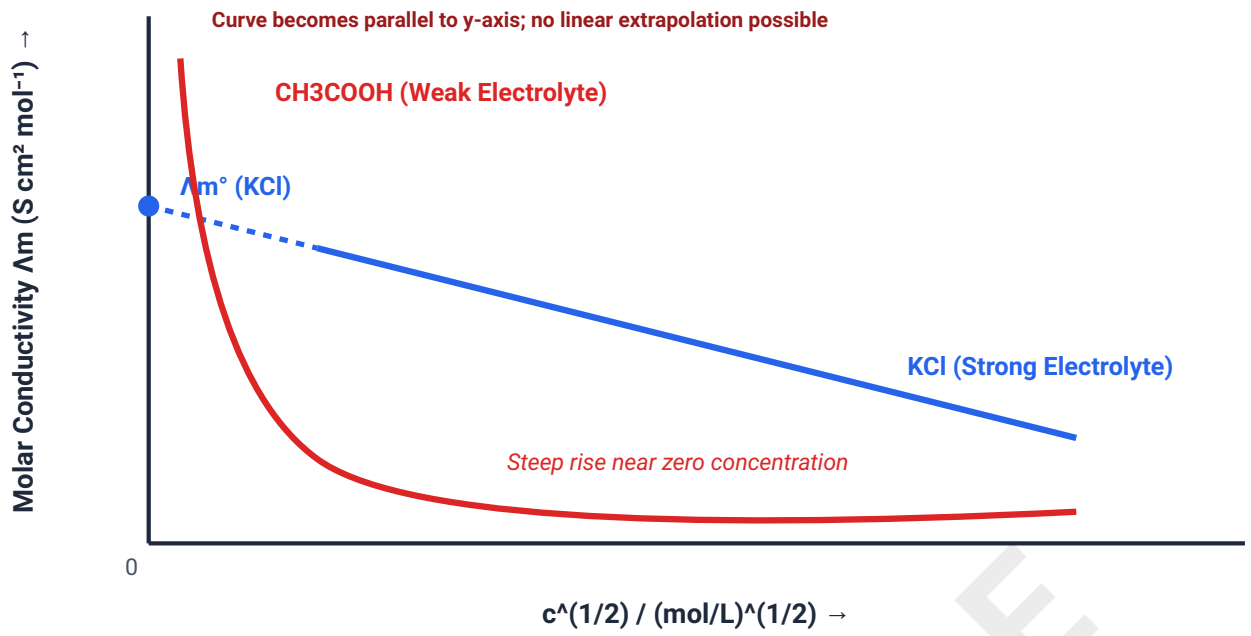
When any solution (whether containing a strong or weak electrolyte) is diluted, the total number of current-carrying ions per unit volume decreases. Consequently, κ **always decreases with decreasing concentration (dilution)**.

6.2 Why Molar Conductivity (Λ_m) Always Increases on Dilution

Molar conductivity is related to conductivity by $\Lambda_m = \kappa V$, where V is the volume of solution containing exactly one mole of dissolved electrolyte. Upon dilution:

- Conductivity κ decreases.
- The volume V containing one mole of electrolyte increases significantly.
- The increase in volume V strongly overcompensates for the decrease in κ. Thus, Λ_m **always increases upon dilution**.

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Tutorial Diagram 5: Plot of molar conductivity (Λ_m) versus $c^{1/2}$ for a strong electrolyte (KCl) and a weak electrolyte ($\mathrm{CH_3COOH}$).

6.3 Strong versus Weak Electrolytes

- **Strong Electrolytes (e.g., KCl, NaCl):** Completely ionized at all realistic concentrations. Dilution merely increases the mean inter-ionic distance, weakening inter-ionic retardations (electrophoretic and relaxation effects). The variation follows the **Debye-Hückel-Onsager equation**:

$$\mathbf{\Lambda_m = \Lambda_m^\circ - Ac^{1/2}}$$

  A linear extrapolation of Λ_m against $c^{1/2}$ to $c \to 0$ yields the intercept Λ_m° (limiting molar conductivity). The slope constant A depends only on the charge stoichiometry of the electrolyte type (e.g., 1-1, 1-2, 2-2) for a given solvent and temperature.
- **Weak Electrolytes (e.g., $\mathrm{CH_3COOH}$):** Dissociate only partially in solution. As dilution proceeds, Ostwald's dilution law drives the degree of dissociation (α) toward 1. At very low concentrations, Λ_m increases sharply, causing the curve to approach the y-axis asymptotically. Therefore, Λ_m° **for weak electrolytes cannot be determined by graphical extrapolation**.

6.4 Kohlrausch's Law of Independent Migration of Ions

Statement: The limiting molar conductivity of an electrolyte at infinite dilution can be represented as the sum of the individual contributions of its constituent cation and anion.

$$\mathbf{\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ}$$

where ν_+ and ν_- denote the number of cations and anions produced per formula unit, and λ_+° and λ_-° represent the limiting molar conductivities of the respective ions.

6.5 High-Yield Applications of Kohlrausch's Law for NEET

1. **Determination of Λ_m° for Weak Electrolytes:**

$$\Lambda_m^\circ(\mathrm{CH_3COOH}) = \Lambda_m^\circ(\mathrm{HCl}) + \Lambda_m^\circ(\mathrm{CH_3COONa}) - \Lambda_m^\circ(\mathrm{NaCl})$$

2. **Calculation of Degree of Dissociation (α):**

$$\alpha = \frac{\mathbf{\Lambda_m}}{\mathbf{\Lambda_m^\circ}}$$

3. **Calculation of Dissociation Constant (K_a):**

$$\mathbf{K_a} = \frac{\mathbf{c\alpha^2}}{\mathbf{1-\alpha}} = \frac{\mathbf{c\Lambda_m^2}}{\mathbf{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)}}$$

7. Electrolytic Cells & Quantitative Laws of Electrolysis

7.1 Faraday's Laws

Faraday's First Law of Electrolysis

The mass (m) of a chemical substance deposited or liberated at any electrode during electrolysis is directly proportional to the quantity of electrical charge (Q) passed through the electrolyte:

$$m \propto Q \implies \mathbf{m = ZQ = ZIt}$$

where I is current in amperes, t is time in seconds, and Z is the **electrochemical equivalent (ECE)**:

$$Z = \frac{\text{Equivalent Weight}}{F} = \frac{\text{Molar Mass}}{n \times 96500}$$

Faraday's Second Law of Electrolysis

When identical quantities of electricity are passed in series through different electrolytic solutions, the masses of different substances liberated at the electrodes are directly proportional to their respective chemical equivalent weights:

$$\frac{\mathbf{m_1}}{\mathbf{m_2}} = \frac{\mathbf{E_1}}{\mathbf{E_2}} \quad \text{where } E = \frac{\text{Molar Mass}}{n}$$

7.2 Competitive Discharge & Products of Electrolysis

When an aqueous solution containing multiple ionic species is electrolyzed, predicting the products requires evaluating reduction potentials, ion concentrations, and activation overpotentials:

Electrolyte System	Cathode Reaction	Anode Reaction	Key Concept / Phenomenon
Molten NaCl	$\mathrm{Na^+ + e^- \rightarrow Na(l)}$	$\mathrm{2Cl^- \rightarrow Cl_2(g) + 2e^-}$	Direct discharge; no competing water molecules.
Aqueous NaCl **(Brine)**	$\mathrm{2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-}$	$\mathrm{2Cl^- \rightarrow Cl_2(g) + 2e^-}$	**Overpotential of** $\mathrm{O_2}$**:** Thermodynamically, water

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Electrolyte System	Cathode Reaction	Anode Reaction	Key Concept / Phenomenon
			oxidation ($E^\circ = 1.23\,\text{V}$) is favored over chloride oxidation ($E^\circ = 1.36\,\text{V}$). However, the oxidation of water to produce O_2 is kinetically very sluggish, requiring an extra voltage (overpotential). Consequently, Cl_2 gas is evolved at the anode.
Dilute H_2SO_4	$2H^+ + 2e^- \rightarrow H_2(g)$	$2H_2O \rightarrow O_2(g) + 4H^+ + 4e^-$	Water oxidation dominates at low acid concentrations ($E^\circ = 1.23\,\text{V}$).
Concentrated H_2SO_4	$2H^+ + 2e^- \rightarrow H_2(g)$	$2SO_4^{2-} \rightarrow S_2O_8^{2-} + 2e^-$	Peroxodisulphate ion formation dominates ($E^\circ = 1.96\,\text{V}$).
Aqueous $AgNO_3$ **(Pt electrodes)**	$Ag^+ + e^- \rightarrow Ag(s)$	$2H_2O \rightarrow O_2(g) + 4H^+ + 4e^-$	Ag^+ has higher reduction potential than water; inert Pt anode.
Aqueous $AgNO_3$ **(Ag electrodes)**	$Ag^+ + e^- \rightarrow Ag(s)$	$Ag(s) \rightarrow Ag^+ + e^-$	**Active electrode:** Anode silver metal dissolves preferentially.

8. Commercial Batteries, Fuel Cells & Corrosion

8.1 Primary Batteries (Non-Rechargeable)

1. Dry Cell (Leclanché Cell)

- **Anode:** Zinc container: $Zn(s) \longrightarrow Zn^{2+} + 2e^-$
- **Cathode:** Carbon (graphite) rod surrounded by powdered MnO_2 and carbon black.
- **Electrolyte:** Moist paste of NH_4Cl and $ZnCl_2$.
- **Cathode Reaction:** $MnO_2 + NH_4^+ + e^- \longrightarrow MnO(OH) + NH_3$
- **Oxidation State Transition:** Manganese is reduced from $+4$ to $+3$.
- Ammonia does not escape as a gas; it forms a complex cation: $[Zn(NH_3)_4]^{2+}$.
- **Cell Potential:** Approximately $1.5\,\text{V}$.

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2. Mercury Button Cell

- **Anode:** Zinc-mercury amalgam ($\mathrm{Zn(Hg)}$).
- **Cathode:** Paste of HgO and carbon.
- **Electrolyte:** Moist paste of KOH and ZnO.
- **Anode Reaction:** $\mathrm{Zn(Hg) + 2OH^- \longrightarrow ZnO(s) + H_2O + 2e^-}$
- **Cathode Reaction:** $\mathrm{HgO(s) + H_2O + 2e^- \longrightarrow Hg(l) + 2OH^-}$
- **Overall Reaction:** $\mathrm{Zn(Hg) + HgO(s) \longrightarrow ZnO(s) + Hg(l)}$
- **NEET High-Yield Fact:** The cell potential is $\approx 1.35\ \mathrm{V}$ and *remains strictly constant* throughout its lifetime because the overall reaction does not involve any dissolved ion whose concentration can change.

8.2 Secondary Batteries (Rechargeable)

1. Lead Storage Battery (Automobile & Inverter Battery)

Electrolyte: 38% aqueous sulphuric acid solution by mass (density $\approx 1.30\ \mathrm{g \cdot mL^{-1}}$).

During Discharge:

- Anode: $\mathrm{Pb(s) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s) + 2e^-}$
- Cathode: $\mathrm{PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \longrightarrow PbSO_4(s) + 2H_2O(l)}$
- **Overall Discharging Reaction:**

$$\mathrm{Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \longrightarrow 2PbSO_4(s) + 2H_2O(l)}$$

During Recharging (External electrical energy supplied):

- Anode process is reversed to regenerate Pb.
- Cathode process is reversed to regenerate $\mathrm{PbO_2}$.
- **Overall Recharging Reaction:**

$$\mathrm{2PbSO_4(s) + 2H_2O(l) \longrightarrow Pb(s) + PbO_2(s) + 2H_2SO_4(aq)}$$

- The density of $\mathrm{H_2SO_4}$ increases during charging and decreases during discharging.

2. Nickel-Cadmium Cell

Has a longer operational life than the lead-acid cell, but is costlier. Overall discharge reaction:

$$\mathrm{Cd(s) + 2Ni(OH)_3(s) \longrightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)}$$

8.3 Hydrogen-Oxygen Fuel Cell

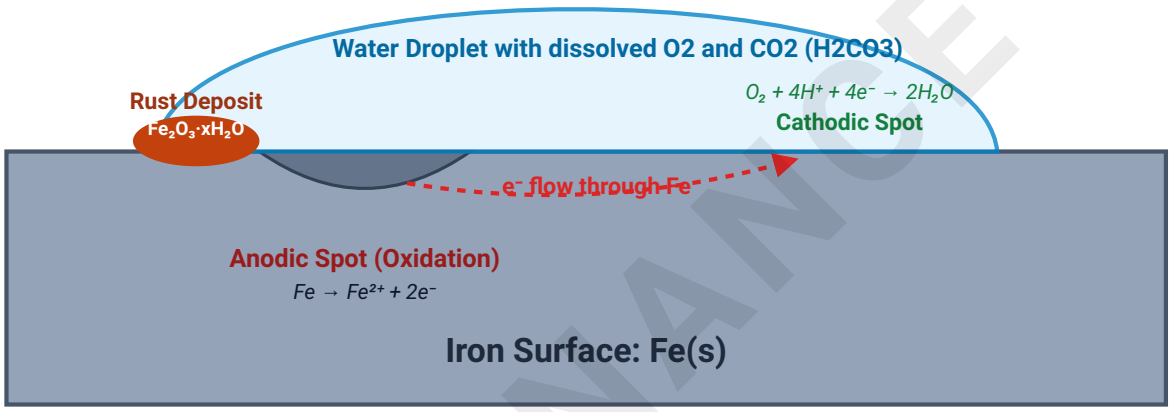
Fuel cells are galvanic cells designed to convert the chemical energy of combustion fuels (e.g., $\mathrm{H_2}$, $\mathrm{CH_4}$, $\mathrm{CH_3OH}$) directly into electrical energy without thermal cycles. Used in the Apollo space missions, where

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product water was consumed by astronauts.

- **Electrodes:** Porous carbon impregnated with catalysts (finely divided Pt or Pd).
- **Electrolyte:** Concentrated aqueous NaOH solution.
- **Anode Reaction:** $2\mathrm{H_2(g)} + 4\mathrm{OH^-(aq)} \longrightarrow 4\mathrm{H_2O(l)} + 4\mathrm{e^-}$
- **Cathode Reaction:** $\mathrm{O_2(g)} + 2\mathrm{H_2O(l)} + 4\mathrm{e^-} \longrightarrow 4\mathrm{OH^-(aq)}$
- **Overall Reaction:** $2\mathrm{H_2(g)} + \mathrm{O_2(g)} \longrightarrow 2\mathrm{H_2O(l)}$
- **Efficiency:** $\approx 70\%$ (substantially higher than thermal power plants, which operate around 40%).

8.4 Corrosion as an Electrochemical Phenomenon



Tutorial Diagram 6: Electrochemical mechanism of iron rusting: oxidation of iron occurs at an anodic site, electrons flow through the metal, and dissolved oxygen is reduced at a cathodic site.

- **Anode (Oxidation):** $2\mathrm{Fe(s)} \longrightarrow 2\mathrm{Fe^{2+}} + 4\mathrm{e^-} \quad (E^\circ = -0.44\,\mathrm{V})$
- **Cathode (Reduction):** $\mathrm{O_2(g)} + 4\mathrm{H^+(aq)} + 4\mathrm{e^-} \longrightarrow 2\mathrm{H_2O(l)} \quad (E^\circ = +1.23\,\mathrm{V})$
- **Overall Cell Potential:** $E^\circ_{\text{cell}} = 1.23 - (-0.44) = \mathbf{+1.67\,V}$
- **Rust Formation:**

$$2\mathrm{Fe^{2+}(aq)} + 2\mathrm{H_2O(l)} + \frac{1}{2}\mathrm{O_2(g)} \longrightarrow \mathrm{Fe_2O_3(s)} + 4\mathrm{H^+(aq)}$$

  Hydrated ferric oxide ($\mathrm{Fe_2O_3 \cdot \mathit{x}H_2O}$) forms rust.
- **Prevention Techniques:** Barrier protection (paint, grease, bisphenol), cathodic/sacrificial protection (coupling to more electropositive metals such as Mg or Zn), and galvanization.

BOSONANCE

9. High-Yield Worked Examples

Worked Example 1: Nernst Equation for a Non-Standard Galvanic Cell

Problem: Calculate the cell EMF at $298\,\mathrm{K}$ for the cell:

$$\mathrm{Mg(s)} \mid \mathrm{Mg^{2+}(0.130\,M)} \parallel \mathrm{Ag^{+}(0.0001\,M)} \mid \mathrm{Ag(s)}$$

Given that $E^\circ_{\text{cell}} = 3.17\,\mathrm{V}$.

Solution:

The cell reaction is:

$$\mathrm{Mg(s) + 2Ag^{+}(aq) \longrightarrow Mg^{2+}(aq) + 2Ag(s)} \quad (n = 2)$$

Applying the Nernst equation:

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.059}{n}\log\frac{[\mathrm{Mg^{2+}}]}{[\mathrm{Ag^{+}}]^2}$$

$$E_{\text{cell}} = 3.17 - \frac{0.059}{2}\log\left(\frac{0.130}{(10^{-4})^2}\right) = 3.17 - 0.0295\log\left(\frac{0.130}{10^{-8}}\right)$$

$$\log(1.30 \times 10^{7}) = \log(1.3) + 7 \approx 0.1139 + 7 = 7.1139$$

$$E_{\text{cell}} = 3.17 - (0.0295 \times 7.1139) = 3.17 - 0.210 = \mathbf{2.96\,V}$$

Worked Example 2: Thermodynamic Quantities and Equilibrium Constant

Problem: For the reaction:

$$\mathrm{2Fe^{3+}(aq) + 2I^{-}(aq) \longrightarrow 2Fe^{2+}(aq) + I_2(s)}$$

the standard cell potential is $E^\circ_{\text{cell}} = 0.236\,\mathrm{V}$ at $298\,\mathrm{K}$. Calculate the standard Gibbs free energy change ($\Delta_r G^\circ$) and the equilibrium constant (K_c).

Solution:

The number of transferred electrons is $n = 2$.

$$\Delta_r G^\circ = -nFE^\circ_{\text{cell}} = -2 \times 96487\,\mathrm{C\cdot mol^{-1}} \times 0.236\,\mathrm{V}$$

$$\Delta_r G^\circ = -45541.86\,\mathrm{J\cdot mol^{-1}} = \mathbf{-45.54\,kJ\cdot mol^{-1}}$$

Using the equilibrium relation:

$$E^\circ_{\text{cell}} = \frac{0.059}{n}\log K_c \implies 0.236 = \frac{0.059}{2}\log K_c$$

$$\log K_c = \frac{0.236 \times 2}{0.059} = \frac{0.472}{0.059} = 8.0$$

$$K_c = 10^{8.0} = \mathbf{1.0 \times 10^{8}}$$

BOSONANCE

Worked Example 3: Determination of Cell Constant, κ, and Λ_m

Problem: The electrical resistance of a conductivity cell filled with $0.1\,\text{mol}\cdot\text{L}^{-1}$ KCl solution is $100\,\Omega$. When filled with $0.02\,\text{mol}\cdot\text{L}^{-1}$ KCl solution, the resistance is $520\,\Omega$. If the conductivity of $0.1\,\text{mol}\cdot\text{L}^{-1}$ KCl is $1.29\,\text{S}\cdot\text{m}^{-1}$, calculate: (i) Cell constant (G^*), (ii) Conductivity of $0.02\,\text{M}$ KCl, and (iii) Molar conductivity of $0.02\,\text{M}$ KCl.

Solution:

(i) Cell Constant:

$$G^* = \kappa \times R = 1.29\,\text{S}\cdot\text{m}^{-1} \times 100\,\Omega = \mathbf{129}\,\text{m}^{-1} = \mathbf{1.29}\,\text{cm}^{-1}$$

(ii) Conductivity of $0.02\,\text{M}$ KCl:

$$\kappa = \frac{G^*}{R} = \frac{129\,\text{m}^{-1}}{520\,\Omega} = \mathbf{0.248}\,\text{S}\cdot\text{m}^{-1} = \mathbf{2.48\times10^{-3}}\,\text{S}\cdot\text{cm}^{-1}$$

(iii) Molar Conductivity:

$$\Lambda_m = \frac{\kappa \times 1000}{M} = \frac{2.48\times10^{-3}\times1000}{0.02} = \mathbf{124}\,\text{S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$$

$$\text{In SI units: } 124\times10^{-4}\,\text{S}\cdot\text{m}^2\cdot\text{mol}^{-1} = \mathbf{1.24\times10^{-2}}\,\text{S}\cdot\text{m}^2\cdot\text{mol}^{-1}$$

10. NEET-UG Exam Practice Questions with Detailed Solutions

Practice Question 1 (Single Correct MCQ)

An external variable opposing potential (E_{ext}) is applied across a standard Daniell cell ($E^\circ = 1.10\,\text{V}$). Which of the following observations is correct when E_{ext} is set to $1.40\,\text{V}$?

A. Electrons continue to flow from the zinc rod to the copper rod, but at an accelerated rate.

B. Current flows from the copper rod to the zinc rod in the external circuit.

C. Zinc dissolves at the zinc electrode and copper deposits onto the copper electrode.

D. Electrons flow from the copper electrode to the zinc electrode, and zinc metal deposits on the zinc rod.

Correct Answer: Option D

Reasoning: When the external opposing potential exceeds the galvanic cell potential ($E_{\text{ext}} > 1.10\,\text{V}$), the cell functions as an **electrolytic cell**. The direction of current and electron flow is reversed: electrons are forced from copper to zinc, causing zinc ions to be reduced and deposited at the zinc electrode ($\text{Zn}^{2+} + 2\text{e}^- \rightarrow \text{Zn}$), while copper dissolves at the copper electrode.

BOSONANCE

Practice Question 2 (Single Correct MCQ)

Given the following standard reduction potentials at $298\ \mathrm{K}$:
$E^\circ_{\mathrm{Cr^{3+}/Cr}} = -0.74\ \mathrm{V}$
$E^\circ_{\mathrm{MnO_4^-/Mn^{2+}}} = +1.51\ \mathrm{V}$
$E^\circ_{\mathrm{Cr_2O_7^{2-}/Cr^{3+}}} = +1.33\ \mathrm{V}$
$E^\circ_{\mathrm{Cl_2/Cl^-}} = +1.36\ \mathrm{V}$
The strongest reducing agent and the strongest oxidizing agent, respectively, are:

A. Cr and $\mathrm{MnO_4^-}$
B. $\mathrm{Mn^{2+}}$ and $\mathrm{Cr^{3+}}$
C. $\mathrm{Cl^-}$ and $\mathrm{Cr_2O_7^{2-}}$
D. $\mathrm{Cr^{3+}}$ and $\mathrm{Cl_2}$

Correct Answer: Option A

Reasoning: The species with the lowest (most negative) reduction potential undergoes oxidation most readily and is therefore the strongest reducing agent (Cr, with $E^\circ = -0.74\ \mathrm{V}$). The species with the highest reduction potential undergoes reduction most readily and is therefore the strongest oxidizing agent ($\mathrm{MnO_4^-}$, with $E^\circ = +1.51\ \mathrm{V}$).

Practice Question 3 (Numerical Application MCQ)

The potential of a hydrogen electrode dipping in a solution of $\mathrm{pH} = 10$ at $298\ \mathrm{K}$ (with $p_{\mathrm{H_2}} = 1\ \mathrm{bar}$) is:

A. $+0.591\ \mathrm{V}$
B. $-0.591\ \mathrm{V}$
C. $0.000\ \mathrm{V}$
D. $-0.0591\ \mathrm{V}$

Correct Answer: Option B

Reasoning: For the half-cell reaction: $\mathrm{H^+(aq) + e^- \longrightarrow \tfrac{1}{2}H_2(g)}$.

$$E = E^\circ - \frac{0.0591}{1}\log\frac{p_{\mathrm{H_2}}^{1/2}}{[\mathrm{H^+}]}$$

Since $E^\circ = 0$ and $p_{\mathrm{H_2}} = 1\ \mathrm{bar}$:

$$E = -0.0591\log\frac{1}{[\mathrm{H^+}]} = -0.0591 \times (-\log[\mathrm{H^+}]) = -0.0591 \times \mathrm{pH}$$

For $\mathrm{pH} = 10$:

$$E = -0.0591 \times 10 = \mathbf{-0.591\ V}$$

BOSONANCE

Practice Question 4 (Statement Evaluation MCQ)

Consider the following statements regarding electrolytic conductance:
Statement I: Specific conductance (conductivity, κ) decreases upon dilution because the number of ions per unit volume of solution decreases.
Statement II: Molar conductance (Λ_m) decreases upon dilution because the ions become farther apart and their ionic mobility decreases.
Choose the correct option:

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

C. Statement I is correct, but Statement II is incorrect.

D. Statement I is incorrect, but Statement II is correct.

Correct Answer: Option C

Reasoning: Statement I is factually correct: conductivity (κ) measures conductance per unit volume ($1\ \text{cm}^3$), and on dilution the number of conducting ions per unit volume decreases. Statement II is incorrect: molar conductivity (Λ_m) *increases* upon dilution because the increase in solution volume containing one mole of electrolyte more than compensates for the decrease in κ.

Practice Question 5 (Kohlrausch's Law MCQ)

The limiting molar conductivities at $298\,\text{K}$ for NaCl, HCl, and CH_3COONa are 126.4, 425.9, and $91.0\,\text{S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$ respectively. If the molar conductivity of a $0.001028\,\text{mol}\cdot\text{L}^{-1}$ acetic acid solution is $48.15\,\text{S}\cdot\text{cm}^2\cdot\text{mol}^{-1}$, its degree of dissociation (α) is:

A. 0.1233

B. 0.0123

C. 0.2466

D. 0.0617

Correct Answer: Option A

Reasoning:

$$\Lambda_m^\circ(\text{CH}_3\text{COOH}) = \Lambda_m^\circ(\text{HCl}) + \Lambda_m^\circ(\text{CH}_3\text{COONa}) - \Lambda_m^\circ(\text{NaCl})$$
$$\Lambda_m^\circ(\text{CH}_3\text{COOH}) = 425.9 + 91.0 - 126.4 = \mathbf{390.5}\,\text{S}\cdot\text{cm}^{\mathbf{2}}\cdot\text{mol}^{\mathbf{-1}}$$

Now calculate α:

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{48.15}{390.5} \approx \mathbf{0.1233}$$

BOSONANCE

Practice Question 6 (Faraday's Laws MCQ)

A steady current of $1.50\,\mathrm{A}$ is passed through an aqueous solution of $\mathrm{CuSO_4}$ for 10 minutes using platinum electrodes. The mass of copper deposited at the cathode is (Atomic mass of $\mathrm{Cu} = 63.5\,\mathrm{u}$, $1\,\mathrm{F} = 96500\,\mathrm{C\cdot mol^{-1}}$):

A. $0.592\,\mathrm{g}$
B. $0.296\,\mathrm{g}$
C. $1.184\,\mathrm{g}$
D. $0.049\,\mathrm{g}$

Correct Answer: Option B

Reasoning: Total charge passed:

$$Q = I \times t = 1.50\,\mathrm{A} \times (10 \times 60\,\mathrm{s}) = 900\,\mathrm{C}$$

Cathodic reduction: $\mathrm{Cu^{2+} + 2e^- \longrightarrow Cu(s)}$.
Deposition of $1\,\mathrm{mol}$ of Cu ($63.5\,\mathrm{g}$) requires $2\,\mathrm{F} = 2 \times 96500\,\mathrm{C} = 193000\,\mathrm{C}$.

$$m = \frac{63.5 \times 900}{193000} \approx \mathbf{0.296\,g}$$

Practice Question 7 (Assertion-Reason MCQ)

Assertion (A): The cell potential of a mercury cell remains constant throughout its useful lifetime.
Reason (R): The overall cell reaction of a mercury cell does not involve any ions in solution whose concentration can change during operation.

A. Both (A) and (R) are true and (R) is the correct explanation of (A).
B. Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
C. (A) is true but (R) is false.
D. (A) is false but (R) is true.

Correct Answer: Option A

Reasoning: The overall reaction is $\mathrm{Zn(Hg) + HgO(s) \longrightarrow ZnO(s) + Hg(l)}$. All reactants and products are either solids, liquid mercury, or amalgam; no ionic concentrations appear in the reaction quotient Q. Thus, by the Nernst equation, E_{cell} is independent of time and electrolyte consumption, maintaining a steady potential of $\approx 1.35\,\mathrm{V}$.

BOSONANCE

Practice Question 8 (Electrolysis Products MCQ)

During the electrolysis of concentrated aqueous NaCl (brine) using inert platinum electrodes, the primary products liberated at the cathode and anode, respectively, are:

A. Na at cathode and $\mathrm{Cl_2}$ at anode
B. $\mathrm{H_2}$ at cathode and $\mathrm{O_2}$ at anode
C. $\mathrm{H_2}$ at cathode and $\mathrm{Cl_2}$ at anode
D. $\mathrm{O_2}$ at cathode and $\mathrm{H_2}$ at anode

Correct Answer: Option C

Reasoning: At the cathode, water ($E^\circ = -0.83\,\mathrm{V}$) is reduced in preference to $\mathrm{Na^+}$ ($E^\circ = -2.71\,\mathrm{V}$), liberating $\mathrm{H_2}$ gas. At the anode, while water oxidation to $\mathrm{O_2}$ is thermodynamically favored ($E^\circ = 1.23\,\mathrm{V}$ vs. $1.36\,\mathrm{V}$), the significant **overpotential of oxygen** makes the oxidation of $\mathrm{Cl^-}$ to $\mathrm{Cl_2}$ gas kinetically preferred.

Practice Question 9 (Lead Storage Battery MCQ)

During the recharging of a discharged lead storage battery, which of the following chemical processes occurs?

A. $\mathrm{PbSO_4}$ deposited on the cathode is reduced to Pb.
B. $\mathrm{PbSO_4}$ deposited on the anode is oxidized to $\mathrm{PbO_2}$.
C. $\mathrm{PbSO_4}$ on the negative plate is converted back to Pb, and $\mathrm{PbSO_4}$ on the positive plate is converted back to $\mathrm{PbO_2}$.
D. Sulphuric acid is consumed and its density decreases.

Correct Answer: Option C

Reasoning: On recharging, current is forced through the battery in reverse. The reaction at the negative plate is $\mathrm{PbSO_4(s) + 2e^- \rightarrow Pb(s) + SO_4^{2-}}$, and at the positive plate is $\mathrm{PbSO_4(s) + 2H_2O \rightarrow PbO_2(s) + SO_4^{2-} + 4H^+ + 2e^-}$. This regenerates $\mathrm{H_2SO_4}$, increasing the acid's density.

Practice Question 10 (Fuel Cell Principles MCQ)

Which of the following statements is **incorrect** regarding the $\mathrm{H_2}$–$\mathrm{O_2}$ fuel cell?

A. It operates with a thermodynamic efficiency of roughly 70%, compared to $\approx 40\%$ for conventional thermal plants.
B. The net chemical reaction is identical to the combustion of hydrogen: $\mathrm{2H_2(g) + O_2(g) \rightarrow 2H_2O(l)}$.
C. Gaseous fuels must be supplied continuously to sustain current output.
D. The electrolyte consumed in the reaction is directly converted to toxic waste, creating environmental disposal challenges.

Correct Answer: Option D

Reasoning: Statement D is incorrect (and thus the correct choice). The only product of the $\mathrm{H_2}$–$\mathrm{O_2}$ fuel cell is pure water, which is completely non-polluting and was even used as drinking water by astronauts on Apollo missions. The alkaline electrolyte is not consumed in the overall reaction.

BOSONANCE

11. Common Pitfalls & NEET Exam Revision Checklist

Frequent Student Traps & How to Avoid Them

- **Sign Error in Reduction Potential:** Remember that reducing power varies *inversely* with reduction potential. A more negative E° means a stronger reducing agent.
- **Molar Conductivity Unit Confusion:** When using κ in $\mathrm{S \cdot cm^{-1}}$, multiply by 1000 to obtain Λ_m in $\mathrm{S \cdot cm^2 \cdot mol^{-1}}$. When κ is in $\mathrm{S \cdot m^{-1}}$, divide by $(1000 \times \mathrm{Molarity})$ to obtain Λ_m in $\mathrm{S \cdot m^2 \cdot mol^{-1}}$.
- **Number of Electrons (n) in the Nernst Equation:** Balance the full redox equation first to identify n. For example, in $\mathrm{Mg + 2Ag^+ \rightarrow Mg^{2+} + 2Ag}$, $n = 2$. Remember to square $[\mathrm{Ag^+}]$ in the reaction quotient Q.
- **Corrosion Reactions:** The anodic site is where iron is oxidized ($\mathrm{Fe \rightarrow Fe^{2+} + 2e^-}$). Reduction of dissolved oxygen occurs at a separate cathodic site on the metal surface ($\mathrm{O_2 + 4H^+ + 4e^- \rightarrow 2H_2O}$), not the other way around.

Rapid NEET Revision Checklist

1. $\Delta_r G^\circ = -nFE^\circ_{\text{cell}} = -2.303RT \log K_c$
2. $E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.059}{n} \log Q$ at $298\ \mathrm{K}$
3. Cell constant $G^* = \frac{l}{A} = R\kappa$
4. $\Lambda_m = \frac{\kappa \times 1000}{M}$
5. Kohlrausch's law: $\Lambda^\circ_m = \nu_+ \lambda^\circ_+ + \nu_- \lambda^\circ_-$
6. Degree of dissociation: $\alpha = \frac{\Lambda_m}{\Lambda^\circ_m}$; Dissociation constant: $K_a = \frac{c\alpha^2}{1-\alpha}$
7. Faraday's 1st Law: $m = ZIt = \left(\frac{M}{nF}\right) It$
8. Dry cell EMF $\approx 1.5\ \mathrm{V}$; Mercury cell EMF $\approx 1.35\ \mathrm{V}$ (constant)
9. Lead storage battery: $\mathrm{H_2SO_4}$ consumed on discharge, regenerated on recharge

BOSONANCE

12. Final Recap

Electrochemistry unifies chemical thermodynamics and redox phenomena. In galvanic cells, spontaneous chemical reactions drive electron flow through an external circuit, producing usable electrical work. In electrolytic cells, an external potential drives non-spontaneous reactions, providing the basis for electroplating and the industrial extraction of reactive metals.

A thorough mastery of the Nernst equation, electrolytic conductance principles, Kohlrausch's law, Faraday's laws of electrolysis, and the chemical reactions of commercial batteries will give students the foundation needed to solve any electrochemistry problem on the NEET UG examination with confidence.